

Limits & Derivatives



Evaluating limits by direct substitution

- 1 Evaluate $\lim_{x \rightarrow 2} (3x^2 - 4x + 1)$.
 - 2 Evaluate $\lim_{x \rightarrow -1} (x^3 + 2x - 5)$.
 - 3 Evaluate $\lim_{x \rightarrow 0} \frac{5x + 7}{x + 3}$.
 - 4 Evaluate $\lim_{x \rightarrow 3} \sqrt{x + 13}$.
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Evaluating limits by factoring (the 0/0 form)

- 5 Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$.
 - 6 Evaluate $\lim_{x \rightarrow -2} \frac{x^2 + 5x + 6}{x + 2}$.
 - 7 Evaluate $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$.
 - 8 Evaluate $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 - 3x + 2}$.
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Limits at infinity

- 9 Evaluate $\lim_{x \rightarrow \infty} \frac{3x^2 + 2x}{x^2 - 5}$.
 - 10 Evaluate $\lim_{x \rightarrow \infty} \frac{2x + 7}{x^2 + 1}$.
 - 11 Evaluate $\lim_{x \rightarrow \infty} \frac{5x^3 - 2}{x^3 + 4x}$.
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One-sided limits

- 12 A function is defined by $f(x) = x + 1$ for $x < 2$, and $f(x) = 2x - 1$ for $x \geq 2$. Find $\lim_{x \rightarrow 2^-} f(x)$.
 - 13 Using the same piecewise function, find $\lim_{x \rightarrow 2^+} f(x)$.
 - 14 Using the same piecewise function, does $\lim_{x \rightarrow 2} f(x)$ exist? State its value.
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The power rule

- 15 Differentiate $f(x) = x^5$.
- 16 Differentiate $f(x) = 3x^4 - 2x^2 + 7$.
- 17 Differentiate $f(x) = \sqrt{x}$.
- 18 Differentiate $f(x) = \frac{1}{x^2}$.
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The product rule

- 19 Differentiate $f(x) = x^2(3x + 1)$ using the product rule.
- 20 Differentiate $f(x) = (x + 2)(x - 5)$ using the product rule.
- 21 Differentiate $f(x) = (2x - 1)(x^2 + 3)$ using the product rule.
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The chain rule

- 22 Differentiate $f(x) = (3x + 1)^4$.
- 23 Differentiate $f(x) = (x^2 - 5)^3$.
- 24 Differentiate $f(x) = \sqrt{4x + 9}$.
- 25 Differentiate $f(x) = (5 - 2x)^6$.
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Derivatives of trigonometric and exponential functions

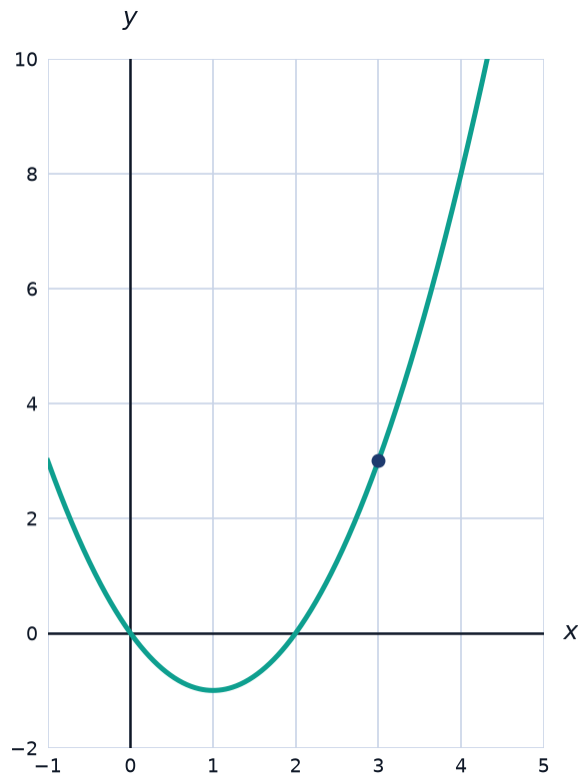
- 26 Differentiate $f(x) = \sin x + \cos x$.
- 27 Differentiate $f(x) = 3\sin x$.
- 28 Differentiate $f(x) = e^x + x^2$.
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Tangent lines

- 29 Find the slope of the tangent line to $f(x) = x^2$ at $x = 3$.

30 Find the equation of the tangent line to $f(x) = x^2$ at $x = 2$.

31 The graph of $f(x) = x^2 - 2x$ is shown. Find the equation of the tangent line at $x = 3$ (the marked point).



32 Find the x -coordinate(s) where the tangent to $f(x) = x^3 - 3x$ is horizontal.

Increasing, decreasing, and the second derivative

33 For $f(x) = x^2 - 4x + 1$, find $f'(x)$ and determine the interval where f is increasing.

34 For $f(x) = -x^2 + 6x$, determine the interval where f is decreasing.

35 Find the second derivative of $f(x) = x^4 - 2x^2$.

Word problems: rates of change

- 36 This question has two parts. The height of a ball thrown upward is given by $h(t) = -5t^2 + 20t + 2$ (in meters, t in seconds).
- a Find the velocity function $h'(t)$.
 - b Find the velocity at $t = 1$ second, and state whether the ball is rising or falling.
- 37 This question has two parts. A company's cost to produce x items is $C(x) = 0.01x^2 + 5x + 200$ (in dollars).
- a Find the marginal cost function $C'(x)$.
 - b Find the marginal cost at $x = 100$ items, and interpret its meaning.
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MCQ — Evaluating limits by direct substitution

- 38 Evaluate $\lim_{x \rightarrow 3} (2x^2 - 5x + 3)$.

- a. 12
- b. -6
- c. 18
- d. 6

- 39 Evaluate $\lim_{x \rightarrow -2} (x^3 - 3x + 4)$.

- a. 10
- b. -2
- c. 2
- d. -10

- 40 Evaluate $\lim_{x \rightarrow 1} \frac{4x + 9}{x + 5}$.

- a. $\frac{9}{5}$
- b. $\frac{13}{6}$
- c. $\frac{13}{5}$
- d. $\frac{4}{5}$

41 Evaluate $\lim_{x \rightarrow 6} \sqrt{x + 10}$.

- a. 4
- b. 8
- c. $\sqrt{16}$
- d. 16

42 Evaluate $\lim_{x \rightarrow 4} \frac{x^2 - 3x}{2x + 1}$.

- a. $\frac{4}{7}$
- b. $\frac{16}{9}$
- c. $\frac{4}{8}$
- d. $\frac{4}{9}$

MCQ — Evaluating limits by factoring (the 0/0 form)

43 Evaluate $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$.

- a. 16
- b. 0
- c. 8
- d. 4

44 Evaluate $\lim_{x \rightarrow -3} \frac{x^2 + 7x + 12}{x + 3}$.

- a. 7
- b. 4
- c. -1
- d. 1

45 Evaluate $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$.

- a. 0
- b. 5
- c. 10
- d. 25

46 Evaluate $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - 5x + 6}$.

- a. -1
- b. $\frac{4}{3}$
- c. -4
- d. 4

47 Evaluate $\lim_{x \rightarrow -1} \frac{x^2 - 1}{x^2 + 3x + 2}$.

- a. -2
- b. 2
- c. -1
- d. $-\frac{1}{2}$

MCQ — Limits at infinity

48 Evaluate $\lim_{x \rightarrow \infty} \frac{4x^2 + 3x}{2x^2 - 7}$.

- a. 2
- b. 0
- c. $\frac{3}{7}$
- d. 4

49 Evaluate $\lim_{x \rightarrow \infty} \frac{3x + 5}{x^2 + 2}$.

a. 3

b. $\frac{5}{2}$

c. ∞

d. 0

50 Evaluate $\lim_{x \rightarrow \infty} \frac{7x^3 - 1}{x^3 + 3x}$.

a. 0

b. 1

c. 7

d. -1

51 Evaluate $\lim_{x \rightarrow \infty} \frac{2x^2 + 1}{5x^3 - 4}$.

a. ∞

b. $\frac{2}{5}$

c. 0

d. 5

MCQ — One-sided limits

52 A function is defined by $f(x) = x + 2$ for $x < 3$, and $f(x) = 3x - 4$ for $x \geq 3$. Find $\lim_{x \rightarrow 3^-} f(x)$.

a. 3

b. 4

c. 9

d. 5

- 53 Using the same piecewise function, find $\lim_{x \rightarrow 3^+} f(x)$.
- a. 3
 - b. 4
 - c. 5
 - d. 9
- 54 Using the same piecewise function, does $\lim_{x \rightarrow 3} f(x)$ exist? State its value.
- a. Yes, it equals 9
 - b. No, it does not exist
 - c. Yes, it equals 5
 - d. Yes, it equals 4
- 55 A function is defined by $f(x) = 2x$ for $x \leq 1$, and $f(x) = x + 3$ for $x > 1$. Find $\lim_{x \rightarrow 1^-} f(x)$.
- a. 1
 - b. 2
 - c. 3
 - d. 4
- 56 Using the same piecewise function, find $\lim_{x \rightarrow 1^+} f(x)$, and state whether $\lim_{x \rightarrow 1} f(x)$ exists.
- a. 4; does not exist
 - b. 4; exists
 - c. 2; exists
 - d. 2; does not exist

MCQ — The power rule

57 Differentiate $f(x) = x^7$.

a. x^6

b. $7x^8$

c. $6x^7$

d. $7x^6$

58 Differentiate $f(x) = 4x^5 - 3x^3 + 2x$.

a. $4x^4 - 3x^2 + 2$

b. $20x^4 - 3x^2 + 2$

c. $20x^4 - 9x^2$

d. $20x^4 - 9x^2 + 2$

59 Differentiate $f(x) = \sqrt{x}$.

a. $\frac{1}{2}\sqrt{x}$

b. $\frac{1}{\sqrt{x}}$

c. $2\sqrt{x}$

d. $\frac{1}{2\sqrt{x}}$

60 Differentiate $f(x) = \frac{1}{x^3}$.

a. $-\frac{3}{x^2}$

b. $-\frac{3}{x^4}$

c. $\frac{3}{x^4}$

d. $-\frac{1}{3x^4}$

61 Differentiate $f(x) = 6x^2 - 4x + 9$.

- a. $6x - 4$
 - b. $12x + 9$
 - c. $12x - 4x$
 - d. $12x - 4$
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MCQ — The product rule

62 Differentiate $f(x) = x^3(2x + 5)$ using the product rule.

- a. $8x^3 + 15x^2$
- b. $6x^2 + 5$
- c. $8x^3 + 5x^2$
- d. $2x^3 + 15x^2$

63 Differentiate $f(x) = (x + 3)(x - 7)$ using the product rule.

- a. $x - 4$
- b. $2x - 4$
- c. $2x - 10$
- d. $2x + 4$

64 Differentiate $f(x) = (3x - 2)(x^2 + 4)$ using the product rule.

- a. $6x^2 - 4x + 12$
- b. $3x^2 - 4x + 12$
- c. $9x^2 + 12$
- d. $9x^2 - 4x + 12$

65 Differentiate $f(x) = (x^2 + 1)(x^2 - 1)$ using the product rule.

- a. $4x^3$
- b. $4x$
- c. $2x^3$
- d. $4x^3 - 2x$

66 Differentiate $f(x) = x^2(x^3 - 4)$ using the product rule.

- a. $5x^4 - 4$
- b. $3x^4 - 8x$
- c. $5x^4 - 8x$
- d. $5x^4 - 8$

MCQ — The chain rule

67 Differentiate $f(x) = (2x + 3)^5$.

- a. $5(2x + 3)^4$
- b. $2(2x + 3)^4$
- c. $10(2x + 3)^5$
- d. $10(2x + 3)^4$

68 Differentiate $f(x) = (x^2 - 3)^4$.

- a. $8x(x^2 - 3)^3$
- b. $4(x^2 - 3)^3$
- c. $4x(x^2 - 3)^3$
- d. $8x(x^2 - 3)^4$

69 Differentiate $f(x) = \sqrt{5x + 1}$.

a. $\frac{1}{2\sqrt{5x + 1}}$

b. $\frac{5}{2\sqrt{5x + 1}}$

c. $5\sqrt{5x + 1}$

d. $\frac{5}{\sqrt{5x + 1}}$

70 Differentiate $f(x) = (4 - 3x)^5$.

a. $-15(4 - 3x)^4$

b. $-5(4 - 3x)^4$

c. $-3(4 - 3x)^4$

d. $15(4 - 3x)^4$

71 Differentiate $f(x) = (x^3 + 2)^2$.

a. $6x^2(x^3 + 2)$

b. $6x^2(x^3 + 2)^2$

c. $3x^2(x^3 + 2)$

d. $2(x^3 + 2)$

MCQ — Derivatives of trigonometric and exponential functions

72 Differentiate $f(x) = \cos x - \sin x$.

a. $\sin x + \cos x$

b. $-\sin x - \cos x$

c. $\sin x - \cos x$

d. $-\sin x + \cos x$

73 Differentiate $f(x) = 5\cos x$.

- a. $5\sin x$
- b. $-5\cos x$
- c. $5\cos x$
- d. $-5\sin x$

74 Differentiate $f(x) = e^x - x^3$.

- a. $e^x - 3x$
- b. $e^x - x^2$
- c. $xe^{x-1} - 3x^2$
- d. $e^x - 3x^2$

75 Differentiate $f(x) = 2e^x + \sin x$.

- a. $2e^x + \cos x$
- b. $2e^x + \sin x$
- c. $e^x + \cos x$
- d. $2e^x - \cos x$

MCQ — Tangent lines

76 Find the slope of the tangent line to $f(x) = x^2$ at $x = 4$.

- a. 4
- b. 2
- c. 16
- d. 8

77 Find the equation of the tangent line to $f(x) = x^2$ at $x = 3$.

a. $y = 9x - 6$

b. $y = 6x + 9$

c. $y = 3x - 9$

d. $y = 6x - 9$

78 The graph of $f(x) = x^2 - 2x$ is shown, with the point at $x = 3$ marked. Find the equation of the tangent line there.



a. $y = 3x - 9$

b. $y = 4x - 3$

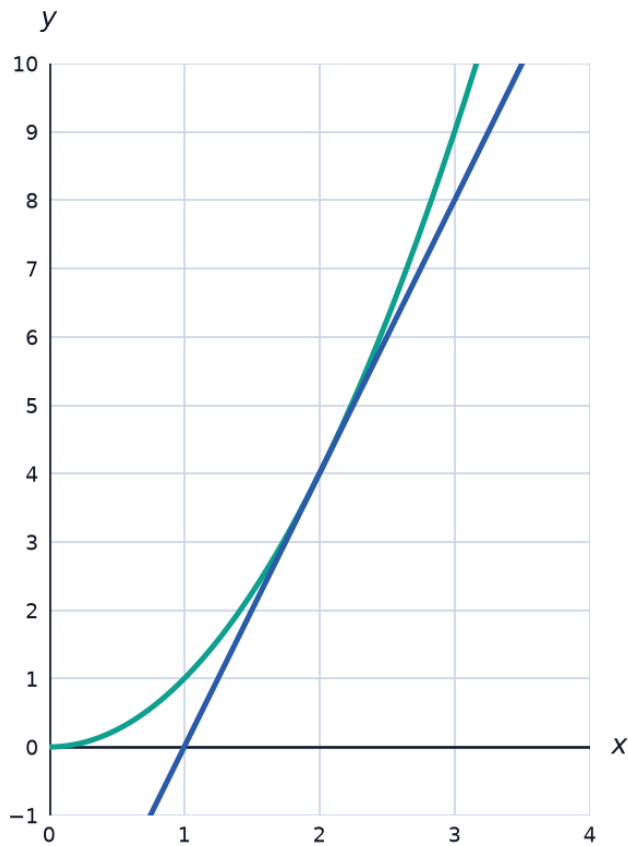
c. $y = 4x + 9$

d. $y = 4x - 9$

79 Find the x -coordinate(s) where the tangent to $f(x) = x^3 - 12x$ is horizontal.

- a. $x = 0$
- b. $x = 4$ and $x = -4$
- c. $x = 2$ only
- d. $x = 2$ and $x = -2$

80 The graph shows $f(x) = x^2$ and its tangent line at $x = 2$. Find the slope of the tangent line by reading the graph.



- a. 4
- b. 2
- c. 1
- d. 8

MCQ — Increasing, decreasing, and the second derivative

81 For $f(x) = x^2 - 6x + 3$, find $f'(x)$ and determine the interval where f is increasing.

- a. $f'(x) = 2x - 6$; increasing for $x < 3$
- b. $f'(x) = 2x - 6$; increasing for $x > 3$
- c. $f'(x) = 2x$; increasing for $x > 3$
- d. $f'(x) = 2x - 6$; increasing for $x > 6$

82 For $f(x) = -x^2 + 8x$, determine the interval where f is decreasing.

- a. $x < 4$
- b. $x > 8$
- c. $x > 4$
- d. $x < 8$

83 Find the second derivative of $f(x) = x^5 - 3x^3$.

- a. $20x^2 - 18x$
- b. $20x^3 - 18x$
- c. $20x^3 - 9x$
- d. $5x^4 - 9x^2$

84 Find the second derivative of $f(x) = x^4 + 2x^2$.

- a. $12x^2$
- b. $4x^3 + 4x$
- c. $12x^2 + 4$
- d. $12x^2 + 4x$

MCQ — Word problems: rates of change

- 85 The height of a ball thrown upward is given by $h(t) = -5t^2 + 30t + 3$ (in meters). Find the ball's velocity at $t = 2$ seconds.
- a. 10
 - b. -10
 - c. 30
 - d. 20
- 86 A company's cost to produce x items is $C(x) = 0.02x^2 + 8x + 150$ (in dollars). Find the marginal cost at $x = 100$.
- a. 8 dollars
 - b. 4 dollars
 - c. 20 dollars
 - d. 12 dollars
- 87 The height of a ball thrown upward is given by $h(t) = -5t^2 + 30t + 3$ (in meters). Find the time at which the ball reaches its maximum height.
- a. $t = 3$
 - b. $t = 5$
 - c. $t = 6$
 - d. $t = 2$

Multiple choice — easy

88 Evaluate $\lim_{x \rightarrow -5} (-3x^2 - 5x + 4)$.

- a. -49
- b. -43
- c. 46
- d. -46

89 Evaluate $\lim_{x \rightarrow -4} (x^2 - x - 2)$.

- a. 18
- b. 21
- c. -18
- d. 15

90 Evaluate $\lim_{x \rightarrow 0} (-2x^2 + x - 2)$.

- a. 1
- b. -2
- c. 2
- d. -5

91 Evaluate $\lim_{x \rightarrow 4} (3x^2 - 7x - 5)$.

- a. 15
- b. 18
- c. 12
- d. -15

92 Evaluate $\lim_{x \rightarrow -1} (2x^2 + 8x - 9)$.

a. -18

b. 15

c. -12

d. -15

93 Differentiate $f(x) = x^5$.

a. x^5

b. $5x^4$

c. $6x^4$

d. $5x^5$

94 Differentiate $f(x) = -2x^3$.

a. $-5x^2$

b. $-2x^3$

c. $-6x^3$

d. $-6x^2$

95 Differentiate $f(x) = -x^3$.

a. $-3x^2$

b. $-x^3$

c. $-3x^3$

d. $-2x^2$

96 Differentiate $f(x) = 4x^3$.

a. $13x^2$

b. $12x^3$

c. $4x^3$

d. $12x^2$

97 Differentiate $f(x) = x^4 - 2x$.

a. $5x^3 - 2$

b. $4x^3 - 2$

c. $4x^4 - 2$

d. $x^4 - 2$

98 Differentiate $f(x) = 4\sin x$.

a. $4\sin x$

b. $-4\cos x$

c. $4\cos x$

d. $5\cos x$

99 Differentiate $f(x) = \sin x + 3x^3$.

a. $\cos x + 10x^2$

b. $\cos x + 9x^2$

c. $\cos x + 3x^3$

d. $\sin x + 9$

100 Differentiate $f(x) = \sin x + 2x^2$.

- a. $\cos x + 4x$
- b. $\cos x + 2x^2$
- c. $\cos x + 5x$
- d. $\sin x + 4$

101 Differentiate $f(x) = \sin x$.

- a. $\cos x$
- b. $-\cos x$
- c. $\sin x$
- d. $-\sin x$

102 Differentiate $f(x) = 3\cos x$.

- a. $-3\sin x$
- b. $3\cos x$
- c. $3\sin x$
- d. $4 - \sin x$

Multiple choice — medium

103 Evaluate $\lim_{x \rightarrow -4} \frac{x^2 + 9x + 20}{x + 4}$.

- a. 1
- b. 3
- c. 4
- d. -1

104 Evaluate $\lim_{x \rightarrow -1} \frac{x^2 - 4x - 5}{x + 1}$.

a. -8

b. -4

c. 6

d. -6

105 Evaluate $\lim_{x \rightarrow 5} \frac{x^2 + x - 30}{x - 5}$.

a. 11

b. 13

c. -11

d. 9

106 Evaluate $\lim_{x \rightarrow -1} \frac{x^2 - 2x - 3}{x + 1}$.

a. 4

b. -4

c. -6

d. -2

107 Evaluate $\lim_{x \rightarrow 5} \frac{x^2 - 8x + 15}{x - 5}$.

a. -2

b. 2

c. 6

d. 4

108 Differentiate $f(x) = (-2x + 1)(x + 2)$ using the product rule.

- a. $4x - 3$
- b. $-4x - 3$
- c. $-4x - 1$
- d. $-4x + 3$

109 Differentiate $f(x) = (3x - 6)(3x - 2)$ using the product rule.

- a. $18x + 24$
- b. $18x - 24$
- c. $-18x - 24$
- d. $18x - 22$

110 Differentiate $f(x) = (-3x - 2)(-x + 5)$ using the product rule.

- a. $6x - 13$
- b. $6x + 13$
- c. $-6x - 13$
- d. $6x - 11$

111 Differentiate $f(x) = (3x + 6)(-2x - 6)$ using the product rule.

- a. $-12x - 30$
- b. $-12x + 30$
- c. $-12x - 28$
- d. $12x - 30$

112 Differentiate $f(x) = (3x - 5)(-2x - 1)$ using the product rule.

- a. $-12x + 9$
- b. $-12x - 7$
- c. $12x + 7$
- d. $-12x + 7$

113 Differentiate $f(x) = (-3x + 4)^6$.

- a. $-18(-3x + 4)^5$
- b. $6(-3x + 4)^5$
- c. $-21(-3x + 4)^5$
- d. $-18(-3x + 4)^6$

114 Differentiate $f(x) = (-3x)^4$.

- a. $4(-3x)^3$
- b. $-12(-3x)^3$
- c. $-12(-3x)^4$
- d. $-15(-3x)^3$

115 Differentiate $f(x) = (-3x - 4)^5$.

- a. $-18(-3x - 4)^4$
- b. $-15(-3x - 4)^5$
- c. $-15(-3x - 4)^4$
- d. $5(-3x - 4)^4$

116 Differentiate $f(x) = (3x)^3$.

a. $12(3x)^2$

b. $9(3x)^3$

c. $9(3x)^2$

d. $3(3x)^2$

117 Differentiate $f(x) = (3x + 1)^5$.

a. $18(3x + 1)^4$

b. $15(3x + 1)^5$

c. $15(3x + 1)^4$

d. $5(3x + 1)^4$

Multiple choice — hard

118 Evaluate $\lim_{x \rightarrow \infty} \frac{4x + 1}{x^2 + 9}$.

a. 9

b. 4

c. 0

d. 1

119 Evaluate $\lim_{x \rightarrow \infty} \frac{3x^2 - 5x}{x^2 - 5}$.

a. 0

b. 4

c. 3

d. $\frac{1}{3}$

120 Evaluate $\lim_{x \rightarrow \infty} \frac{4x^2 + 8x}{3x^2 - 8}$.

a. 0

b. $\frac{7}{3}$

c. $\frac{3}{4}$

d. $\frac{4}{3}$

121 Evaluate $\lim_{x \rightarrow \infty} \frac{5x - 7}{x^2}$.

a. 1

b. 0

c. 8

d. 5

122 Evaluate $\lim_{x \rightarrow \infty} \frac{4x^2 - 8x}{2x^2 - 8}$.

a. 2

b. $\frac{1}{2}$

c. 3

d. 0

123 Find the equation of the tangent line to $f(x) = -3x^2 - 4x - 4$ at $x = 1$.

a. $y = 10x - 1$

b. $y = -10x + 2$

c. $y = -10x + 1$

d. $y = -10x - 1$

124 Find the equation of the tangent line to $f(x) = 2x^2 + 3x - 1$ at $x = -3$.

- a. $y = 9x - 19$
- b. $y = -9x + 19$
- c. $y = -9x - 16$
- d. $y = -9x - 19$

125 Find the equation of the tangent line to $f(x) = -3x^2 + 5x + 1$ at $x = 1$.

- a. $y = -x - 4$
- b. $y = -x + 7$
- c. $y = x + 4$
- d. $y = -x + 4$

126 Find the equation of the tangent line to $f(x) = -x^2 - 3x + 2$ at $x = 2$.

- a. $y = 7x + 6$
- b. $y = -7x - 6$
- c. $y = -7x + 9$
- d. $y = -7x + 6$

127 Find the equation of the tangent line to $f(x) = -2x^2 - 4x - 1$ at $x = -2$.

- a. $y = -4x + 7$
- b. $y = 4x - 7$
- c. $y = 4x + 7$
- d. $y = 4x + 10$

128 For $f(x) = 2x^2 + 8x - 4$, determine the interval where f is decreasing.

- a. $x < -2$
- b. $x > -2$
- c. $x < 0$
- d. $x > 0$

129 For $f(x) = -x^2 - 3$, determine the interval where f is increasing.

- a. $x < 0$
- b. $x > 2$
- c. $x < 2$
- d. $x > 0$

130 For $f(x) = x^2 + 4x + 9$, determine the interval where f is decreasing.

- a. $x > 0$
- b. $x < -2$
- c. $x > -2$
- d. $x < 0$

131 For $f(x) = -2x^2 - 4x - 5$, determine the interval where f is increasing.

- a. $x > -1$
- b. $x < -1$
- c. $x < 1$
- d. $x > 1$

132 Find the second derivative of $f(x) = -2x^4 - 5x^2$.

a. $-22x^2 - 10$

b. $24x^2 - 10$

c. $-24x^2 - 10$

d. $-24x^2 + 10$